CHAKRAVYUH / GLUCOSE TWIN

Architecture and data boundaries

OFFLINE EXPERIMENT

Synthetic generator
Seed 2026 / 90 patients
Static EHR + wearable CSV
7 days / 15-minute samples

→

Patient-level split
54 train / 18 validation / 18 test
Past-window features only
Future CGM used as labels

→

Fit and validate
Ridge + logistic regression
Train-only standardization
Validation threshold + intervals

→

Export artifacts
model.json
metrics.json
6 test patients in data.js

BROWSER RUNTIME

Virtual patient state
Static EHR for selected patient
CGM + activity + sleep history
Replay cursor bounds history

→

Quality and feature gate
12 readings / 3-hour window
Withhold if latest CGM absent
Withhold if >3 values absent

→

Local model inference
8 glucose forecasts to +120 min
Excursion score for >180 mg/dL
Empirical marginal intervals

→

Doctor dashboard
Trajectory + review queue
Scenario sensitivity
Local JSON case export

Exported coefficients load into the browser engine. Future readings are excluded at the replay cursor.

Scope and safeguards
All patient data is synthetic. No live EHR, wearable account, backend database, or medication decision engine.
Python and JavaScript feature parity is tested. Whole patients remain in one split; overlapping windows are correlated.
Scenario changes show model sensitivity, not causal treatment effects. Test results do not establish clinical validity.

Source: src/pipeline.py, dashboard/engine.js, dashboard/app.js and model/metrics.json